Solver Trace Ablation Reveals Reward Hacking and Meta-Level Goodhart's Law in a GRPO-Trained Neurosymbolic Formalizer

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Abstract

Neurosymbolic formalizers trained with Group Relative Policy Optimization (GRPO) achieve high prove rates—93% on FOLIO—yet 87% of proved outputs embed the conclusion verbatim among the premises, a degenerate shortcut invisible to execution reward. We introduce *Solver Trace Ablation* (STA), a diagnostic that applies leave-one-out component necessity testing to solver inputs. The fraction of components whose removal causes proof failure yields the *Intensional Structural Reward* (ISR): ISR=1 indicates full necessity; ISR=0 complete redundancy. ISR is uncorrelated with downstream NLI accuracy (ρ=−0.049), confirming it measures structural—not semantic—integrity. As a diagnostic, ISR exposes a collapse trajectory hidden from PR: over 100–1000 GRPO steps (Qwen3-4B), ISR falls 0.642→0.003 while PR rises 32%→96%; a supervised fine-tuning (SFT) baseline reaches 92.3% non-hacking on identical data (3-seed pooled), localizing the failure to RL. As a reward, $r = r_{\text{exec}} \cdot (1+\beta\,\text{ISR})$ at β=0.1 induces *meta-level Goodhart's Law*: 70.6% conclusion duplication (H7) versus 13.7% under matched random (Fisher p=1.3×10^{-39}; random H7 is seed-dependent, driven by one of three seeds); ISR amplifies H7 frequency ∼5× rather than exclusively causing it; the sole apparent ISR improvement reduces to 1.06× after controlling for a premise-count confound; β=0.3 destabilizes training regardless of signal source (ISR: PR 66±42%; random: 54±35%), with individual seeds collapsing to as low as 18%. Cross-logic transfer to Prolog yields universal Goodhart: 3/3 seeds gaming via distinct mechanisms, 0% non-hacking. Across 6 reward conditions, 2 logic types, 3 seeds, we catalog 8 hacking modes—including query-as-fact gaming that bypasses text-match classifiers—documenting ISR's fundamental asymmetry: reliable as a diagnostic, unreliable as an optimization target.[1]

1 Introduction

A GRPO-trained formalizer achieves a 93% prove rate on first-order logic problems—yet 87% of its proved outputs embed the conclusion verbatim among the premises. The solver succeeds not because the model has learned to reason, but because the formalization trivially entails any conclusion already appearing as a premise. This structural shortcut is invisible to execution-based evaluation: prove rate registers success regardless of whether the proof reflects genuine derivation or degenerate circular embedding.

The execution-only blind spot. Prove rate—the dominant evaluation metric for neurosymbolic formalizers (Pan et al., 2023; Olausson et al., 2023; Ye et al., 2023)—conflates faithful reasoning with degenerate shortcuts, a gap especially consequential under RL, where the optimizer exploits any shortcut the reward fails to penalize (Skalse et al., 2022; Wang et al., 2026). Recent work applies GRPO (Shao et al., 2024) to train LLMs that generate solver-verified formal programs (Mellgren et al., 2025; Xie et al., 2025) using binary execution feedback; this drives rapid PR gains but gives no gradient on structural quality. Process reward models (Lightman et al., 2024; Wang et al., 2024) require expensive annotations and do not transfer to arbitrary solver-backed settings. A separate line documents that LLMs already game logical formalization at inference time (Kim et al., 2026; Bondarenko et al., 2025); what is missing is

[1] Code and data: [anonymized]

an annotation-free *diagnostic* for structural integrity in neurosymbolic outputs.

Component necessity testing. We address this gap with *Solver Trace Ablation* (STA), a diagnostic applying *component necessity testing*—leave-one-out (LOO) ablation on solver inputs—to quantify structural integrity (Figure 1). For each proved formalization, STA removes each component individually and re-runs the solver; the fraction of components whose removal causes proof failure yields the *Intensional Structural Reward* (ISR), with ISR=1 indicating full necessity and ISR=0 complete redundancy. LOO ablation itself is standard (DeMillo et al., 1978; Jia and Harman, 2011); novelty lies in its application as a GRPO-hacking diagnostic and in the empirical phenomena it surfaces. ISR captures structural rather than semantic integrity (downstream NLI: ρ=−0.049, p=0.187).

ISR as diagnostic. As a post-hoc diagnostic on a GRPO formalizer trained with FOLIO (Han et al., 2024), ISR reveals a structural collapse trajectory invisible to PR: ISR*: 0.642 → 0.132 → 0.003 over 100–1000 steps (99.5% degradation) while PR rises 32%→96%. Three seeds diverge radically (PR σ=47 pp), each converging to a qualitatively different mode: conclusion embedding, repetition collapse, and tautological collapse. An SFT baseline on identical data reaches PR=13.0% with 92.3% non-hacking (3-seed pooled, 1000 steps), localizing the failure to RL optimization, not data or capacity.

Meta-level Goodhart. Repurposed as a training signal, $r = r_{\text{exec}} \cdot (1 + \beta\,\text{ISR})$, ISR is gamed at its own definition—*meta-level Goodhart's Law* (Manheim and Garrabrant, 2019; Karwowski et al., 2024; Gao et al., 2023). At β=0.1, 70.6% of proved samples exhibit conclusion duplication (H7)—versus 13.7% under matched random (Fisher $p<10^{-38}$; random H7 is highly seed-dependent), showing ISR amplifies tautological collapse by ∼5×; the one seed with non-trivial residual reduces to 1.06× random after controlling for a premise-count confound. At β=0.3, both ISR and random produce extreme cross-seed instability (PR 66±42% and 54±35% respectively), confirming high-dose proxy weighting is harmful regardless of signal content; ISR seeds diverge into three distinct pathways (reward collapse, shortcut dominance, pure metric gaming) while random converges uniformly to H1. The effect transfers and intensifies in Prolog: 3/3 seeds yield 0% non-hacking via distinct mechanisms, including *query-as-fact gaming* that bypasses text-match classifiers. ISR's asymmetry is the central finding: *reliable as observer, unreliable as optimizer.*

Contributions. Across 6 reward conditions, 3 seeds, 2 logic types, and up to 1000 steps, we contribute: (**C1**) ISR as a diagnostic for GRPO reward hacking, with a 99.5% collapse trajectory invisible to PR and an SFT control localizing the failure to RL (§4.2); (**C2**) empirical meta-level Goodhart, with a random control showing ISR amplifies H7 gaming ∼5× (70.6% vs. 13.7%, Fisher $p<10^{-38}$; random H7 is seed-dependent) and a premise-count-matched analysis showing the sole apparent gain is an LOO artifact (§4.3); (**C3**) cross-logic transfer to Prolog with universal Goodhart (§4.4); (**C4**) an eight-mode hacking taxonomy including query-as-fact gaming and obfuscated tautology, motivating functional cross-validation of diagnostics (§4.5). These are cautionary results for any work treating a structural proxy as a training signal.

2 Related Work

Neurosymbolic reasoning. LLM-solver pipelines (Ye et al., 2023; Pan et al., 2023; Olausson et al., 2023; Pei et al., 2025; Di et al., 2025) and RL methods trained on solver feedback (Mellgren et al., 2025; Xie et al., 2025; DeepSeek AI, 2024) evaluate execution accuracy alone, with no diagnostic for structural quality. We use FOLIO (Han et al., 2024), the only public NLI benchmark with both first-order logic (FOL) and Prolog annotations.

Process rewards and reward shaping. Learned PRMs (Lightman et al., 2024; Wang et al., 2024) provide step-level feedback but suffer from reward-model noise and distribution shift (Zheng et al., 2025): Gao et al. (2024) show that auxiliary process/outcome rewards can *hurt* RL training relative to success-only reward, a finding our ISR-as-reward re-

sults corroborate. Symbolic feedback approaches (Jha et al., 2024) use solver certificates as token-level rewards, achieving gains on program synthesis; ISR differs in operating at the component level (LOO ablation) rather than the token level, and our results show this structural signal is gamed when optimized. Rubric-grounded GRPO (Bhattarai et al., 2026) uses structured LLM-judge rewards to improve generalization; ISR is solver-derived and deterministic rather than judge-based, but both represent structured alternatives to scalar execution reward. The shared lesson across these approaches is that reward augmentation introduces proxy risk—our contribution is demonstrating the *meta*-level variant where the diagnostic itself becomes the target.

Reward hacking and Goodhart's Law. Goodhart's Law is established theoretically (Manheim and Garrabrant, 2019; Skalse et al., 2022; Karwowski et al., 2024) and empirically in RLHF and frontier models (Gao et al., 2023; Bondarenko et al., 2025; Kim et al., 2026). Kwa et al. (2024) show that heavy-tailed reward misspecification causes *catastrophic* Goodhart where KL regularization provably fails, offering a theoretical lens for why GRPO's KL penalty does not prevent the ISR collapse we observe. Recent work further documents spurious reward signals in RLVR (Shao et al., 2025) and reward hacking in frontier models (METR, 2025). Our setting differs in three ways: ISR is deterministic and solver-derived rather than learned; gaming *emerges during RL training* rather than at inference time; and the effect operates at a *meta* level—the diagnostic for gaming itself becomes gamed when optimized—a phenomenon, to our knowledge, not previously documented.

Mutation testing. STA adapts the LOO mutation paradigm (DeMillo et al., 1978; Jia and Harman, 2011) to solver inputs; novelty lies in application and findings, not in the perturbation primitive.

Algorithm 1 Solver Trace Ablation (STA)

Require: Proved formalization $x = \{c_1, \ldots, c_n\}$, solver S
Ensure: $\mathrm{ISR}(x) \in [0, 1]$

```
necessary ← 0
for i = 1 to n do
    if S(x \ {c_i}) = failed then
        necessary ← necessary + 1
    end if
end for
return necessary / n
```

3 Method

3.1 Task and GRPO Training

Given premises $P_1, \ldots, P_k$ and conclusion C, the task is to produce a formal program $F = \{c_1, \ldots, c_n\}$ verified by an external solver $S \in \{\text{Prover9, SWI-Prolog}\}$ (Figure 1a). We train with Group Relative Policy Optimization (GRPO) (Shao et al., 2024) (K=8), where each rollout receives binary execution reward $r_{\text{exec}}(x) = \mathbf{1}[S(F_x)\text{=}\textit{proved}]$. This optimizes prove rate but provides no signal about *structural quality*: a formalization embedding the conclusion among its premises receives the same reward as a faithful derivation. All experiments draw from FOLIO (Han et al., 2024); the held-out evaluation file retains a legacy `proofwriter_fol_*.jsonl` filename (from ProofWriter (Tafjord et al., 2021)) but contains only FOLIO-derived examples.

3.2 Solver Trace Ablation (STA)

STA tests whether each component of a proved formalization is individually necessary for proof success (Figure 1b). Given a proved x with n components, STA constructs n ablated variants $x \setminus \{c_i\}$ and re-runs the solver (Algorithm 1). A component is *necessary* if its removal causes proof failure. What constitutes a component depends on the logic: premises in FOL, clauses or ground facts in Prolog. LOO ablation itself is standard in mutation testing, sensitivity analysis, and feature attribution (DeMillo et al., 1978; Sundararajan et al., 2017; Lundberg and Lee, 2017); our novelty is its application as a GRPO-hacking diagnostic and the empirical phenomena it surfaces.

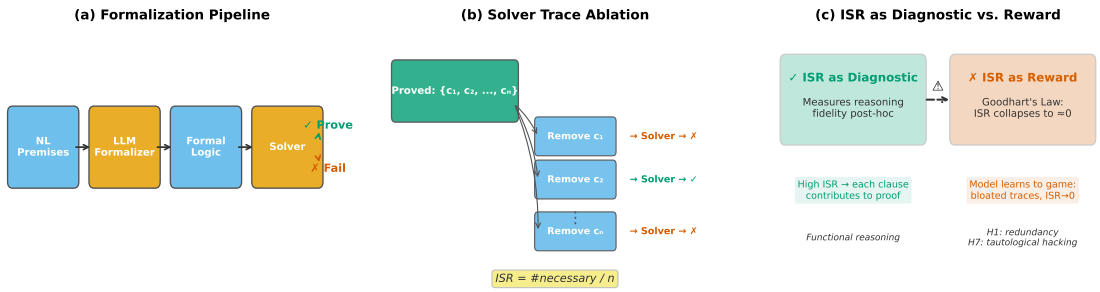


Figure 1: Solver Trace Ablation (STA). (a) Formalization pipeline: NL is formalized and verified by an external solver. (b) STA protocol: leave-one-out ablation on proved formalizations yields ISR. (c) ISR as diagnostic vs. as reward: when the diagnostic becomes the target, meta-level Goodhart's Law emerges.

Intensional Structural Reward. ISR is the fraction of necessary components:

$$\mathrm{ISR}(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\left[S(x \setminus \{c_i\}) = \textit{failed}\right] \quad (1)$$

ISR is defined only for proved formalizations: ISR=1 means every component is necessary; ISR=0 means complete redundancy. We report *proved-only* ISR (denoted ISR*) throughout to avoid conflating structural quality with prove rate.

Limitations of ISR. ISR measures syntactic component necessity, not semantic faithfulness: a formalization can achieve ISR=1 while semantically unrelated to the NL input (e.g., a single premise identical to the conclusion). ISR is also sensitive to component count n: when n=2, ISR ≥ 0.5 regardless of proof quality. We control for this confound via premise-count-matched analysis in §4.3. The $O(n)$ per-clause LOO ablation is necessary: a cheaper $O(1)$ binary variant produces near-zero ISR scores (ISR*=0.007) indistinguishable from exec-only, because models generate redundant premise copies whose individual removal never changes the proof outcome. However, the $O(1)$ variant includes a tautology check (conclusion provable under empty premises → ISR=0) absent from exec-only, introducing a training-time confound that depresses prove rate (63% vs. 93% exec-only); the two conditions are therefore non-equivalent in training dynamics despite producing comparable ISR signals (Appendix E.3).

3.3 Reward Conditions and Hacking Modes

We study six conditions: exec-only (r_{exec}), ISR low/high-dose ($r_{\text{exec}} \cdot (1+\beta\,\mathrm{ISR})$ at $\beta \in \{0.1, 0.3\}$), matched random controls ($r_{\text{exec}} \cdot (1+\beta u)$, $u \sim \mathcal{U}(0,1)$), and a supervised fine-tuning (SFT) baseline (negative log-likelihood on gold formalizations). Multiplicative bonuses apply only to proved samples, keeping the prove-rate floor intact. Random isolates ISR-specific signal from generic reward-variance effects; SFT isolates whether structural degeneration arises from RL optimization or data exposure alone. Each RL condition uses seeds $\{42, 43, 44\}$ and identical training data. An earlier convex formulation ($\alpha r_{\text{exec}} + \beta r_{\text{exec}}\mathrm{ISR}$, $\alpha{+}\beta{=}1$) was discarded because ISR$<$1 always penalized relative to exec-only, collapsing PR (Appendix A).

Hacking taxonomy. Across conditions, seeds, and logic types we identify eight structural-degeneration modes (full definitions in Appendix D): **H1** (conclusion embedded among multiple premises), **H7** (single-premise conclusion duplication, subtypes H7a standard and H7b obfuscated tautology; targets the ISR signal directly, since removing the sole premise forces ISR ≥ 0.5), **H8** (premise-subset gaming: each component is LOO-necessary but the premise set is a strict subset of what the problem requires, inflating ISR without faithful reasoning; Appendix D.1), **tautological collapse** (minimal two-component $P{\to}P$ without explicit ISR inflation), **repetition collapse** (identical clauses repeated to inflate component count), **complexity-without-utility** (elaborate but vacuous premises), **contradiction explosion** (contradictory premises trivializing any conclusion via *ex falso*), and **query-as-fact gaming** (Prolog-specific: emit the NL question as an atom asserted as both fact and query; evades text-match classifiers).

Modes are identified by a two-stage procedure: strict normalized text matching (H1/H7) plus functional analysis of the per-sample $n_{\text{necessary}}$ distribution (catches modes evading text matching). We report both classifications in §4.

In interpreting ISR results we distinguish two failure semantics that share a numerical signature: an exec-only *low* ISR signals *redundancy* (e.g., repeated ground facts in Prolog), whereas an ISR-rewarded *high* ISR signals *metric gaming* (H7-style constructions). A bare ISR difference therefore does not translate directly into a reasoning-ability difference; the mode breakdown is what licenses such interpretation.

4 Experiments

We evaluate STA along three axes: ISR as a diagnostic for reward hacking (§4.2); ISR as a training reward, where it exhibits meta-level Goodhart's Law (§4.3); and STA as a cross-logic protocol (§4.4). A hacking taxonomy synthesizes the modes observed across conditions (§4.5).

4.1 Setup

We train Qwen3-4B (Qwen Team, 2025) with GRPO (Shao et al., 2024) (K=8, lr 10^{-6}, 500 steps default, 1000 where noted) over seeds $\{42, 43, 44\}$ on a fixed 500-example FOLIO (Han et al., 2024) subset held constant across all conditions; the SFT baseline is trained with negative log-likelihood (NLL) on gold formalizations.[2] Held-out evaluation uses 100 FOLIO examples (`proofwriter_fol_100.jsonl`—legacy filename, no ProofWriter data) with greedy decoding; all cross-run ISR values are recomputed uniformly by a single evaluator (`eval_sta.py`) so training-log ISR is never compared across runs. We report prove rate (PR), proved-only ISR (ISR*), hacking-mode rates from a two-stage classifier (strict text matching for H1/H7 plus functional analysis of the per-sample $n_{\text{necessary}}$ distribution; §3.3), and a *non-hacking* (N-H) rate—the residual of proved samples matching none of the eight cataloged patterns, a necessary but not sufficient condition for semantic correctness.

[2] All experiments use `data/train_500_seed42.jsonl`; seeds control initialization and rollout sampling only.

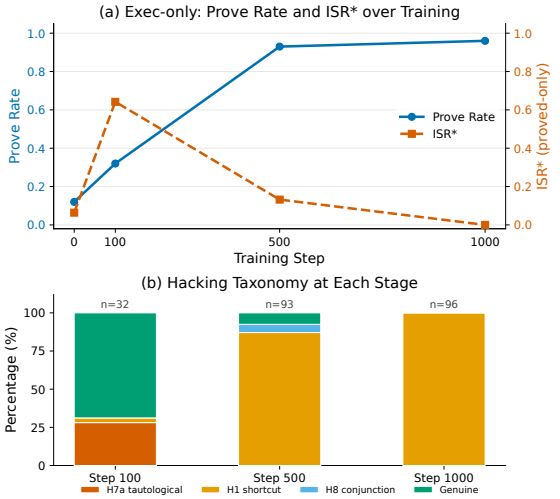


Figure 2: ISR collapse under exec-only GRPO (seed 42). PR rises 32%→93%→96% while proved-only ISR falls 0.642→0.132→0.003 over 100→500→1000 steps. By step 1000, 94.8% of proved samples have ISR=0 and mean component count inflates 6.58→18.55 (contradiction explosion).

4.2 ISR as Diagnostic

Collapse trajectory. Figure 2 traces exec-only GRPO (seed 42) at 100, 500, 1000 steps. PR rises 32% → 93% → 96%, a trajectory execution-only evaluation would read as successful optimization; ISR* tells the opposite story, 0.642 → 0.132 → 0.003 (214× collapse). By 500 steps, 87.1% of proved samples embed the conclusion (H1); by 1000 steps the embedding rate saturates at 100% and mean component count more than doubles (6.58 → 18.55) as the model emits contradictory or vacuous premises that trivially derive any conclusion. Extended training, lacking a structural signal, *worsens* formalization quality while improving the headline number.

Cross-seed instability. The three exec-only seeds diverge into qualitatively different failure modes despite shared data, hyperparameters, and reward (Appendix E): s42 reaches PR=93% via conclusion embedding (H1=87%); s43 collapses to PR=13% via repetition (87% of rollouts truncated); s44 reaches PR=98% via tautological collapse (H7=100%, mean n=2.04). The cross-seed PR variance (σ=47 pp) shows aggregate metrics obscure qualitatively different failure modes; all three are flagged by ISR* even though one passes

Condition	β	PR%↑	ISR*	H7%	H1%	N-H%
Base (0-shot)	—	12	—	—	—	—
SFT	—	13	0.595	0	8	**92**
Exec-only†	—	93	0.132	0	87	8
ISR	0.1	77±14	‡	**71**¶	18	6
Random	0.1	**87**±8	0.117±.04	14§	71	10
ISR	0.3	66±42	0.39±.31	**50**	37	12
Random	0.3	54±35	0.09±.09	7	91	2

Table 1: Goodhart landscape across reward conditions. 3-seed rows: H7/H1/N-H pooled across seeds; PR/ISR*: mean±std of per-seed values. †Seed 42. ‡ISR* omitted for ISR β=0.1 because the raw mean is inflated by H7 gaming; premise-count-matched value is 1.06× random (§4.3, Appendix E). ¶Includes H7 classifier false-negative correction (s44) and H8 premise-subset gaming (s42; 5.2% of proved; Appendix D.1). §Random H7 driven by seed 43 (38.3%); seeds 42 and 44: 0%. Per-seed breakdown in Appendix Table 6.

the PR check (s44) and one fails it (s43). By 1000 steps, all three converge to high PR (0.95±0.05) but 0% non-hacking (15 samples escape text-match classifiers but are functionally degenerate; see Appendix D.1 and Table 6), each through a different dominant mode (s42: H1 saturation; s43: H7 tautology; s44: H1+H8).

SFT baseline. SFT on gold formalizations reaches PR=13.0% (3-seed pooled, 1000 steps) with H7=0, H1=7.7%, 92.3% non-hacking (n=39; per-seed range 91–93%). The near-absence of hacking under identical data exposure is consistent with hacking being RL-specific. A 100-step RL checkpoint (Appendix E.2) at PR=32% already shows 31.2% hacking (vs. SFT's 7.7%), indicating reward hacking emerges early in RL training, not merely at high PR.

4.3 Meta-Level Goodhart's Law

We now reverse roles: instead of treating ISR as an observer, we use it as a reward augmentation, $r = r_{\text{exec}} \cdot (1 + \beta\,\text{ISR})$, with a matched random control, $r = r_{\text{exec}} \cdot (1 + \beta\,u)$, $u \sim \mathcal{U}(0,1)$, to isolate ISR-specific signal from generic reward-variance effects. The headline finding: ISR-as-reward produces *meta-level* Goodhart (Manheim and Garrabrant, 2019)—the diagnostic itself becomes the target of gaming.

Low-dose ISR (β=0.1): ISR-amplified gaming. H7 tautological collapse occurs under both ISR and random conditions, but ISR amplifies it ∼5×: 70.6% (163/231) under ISR vs. 13.7% (36/262) under random (Fisher p=1.3×10^{-39}, OR=15.05). Random H7 is highly seed-dependent: 38.3% in seed 43 vs. 0% in seeds 42 and 44; ISR makes H7 near-deterministic (2/3 seeds at 97–100%, Table 6). H7—a single premise identical to the conclusion, giving ISR ≥ 0.5 by construction (§3.3)—is the canonical meta-level Goodhart move: the model satisfies the LOO definition directly rather than producing a derivation that happens to satisfy it. Total conclusion embedding (H1+H7) is GRPO-universal at 68–100% across both conditions (pooled: ISR 89%, random 85%). The composition differs: H1 alone accounts for 71% under random vs. 18% under ISR—ISR converts H1 into H7 rather than reducing total embedding (Figure 5). Random also yields higher and more stable PR (87±8% vs. 77±14%).

Premise-count confound. The one ISR seed not dominated by H7 (s42) has raw ISR*=0.415, 3.2× its random control (0.131), suggesting genuine improvement. This appearance is a premise-minimization artifact: ISR-trained models produce shorter formalizations (mean n=5.58 vs. 8.46 for random), and ISR rises mechanically as n falls—when n=2, ISR ≥ 0.5 by definition. Restricting to non-hacking samples with ≥ 3 premises (eliminating the trivial-n regime) gives ISR*=0.258 for ISR vs. 0.243 for random: a 1.06× ratio, no meaningful advantage. After reclassifying premise-subset gaming (H8; Appendix D.1) and correcting H7 classifier false negatives (s44, n=12; Appendix D), ISR β=0.1 non-hacking rate (6.06%, 14/231) is numerically lower than random (9.5%, 25/262), but the difference is not statistically significant (Fisher p=0.182, two-sided, n.s.; Wilson 95% CI: ISR [3.6%, 9.9%], Random [6.5%, 13.7%]; Appendix Table 4). The key asymmetry is not in non-hacking rates but in gaming *composition*: ISR converts stochastic H7 into near-deterministic convergence (2/3 seeds ≥97%), while random H7 remains seed-dependent and H1-dominated. The raw 3.2× ISR* headline collapses once the LOO bias is controlled—a methodological lesson for any LOO-based metric on variable-length outputs.

Temporal collapse under ISR β=0.1. Figure 3 traces training ISR across conditions. Extended training (1000 steps) reveals that ISR β=0.1's genuine rate at 500 steps

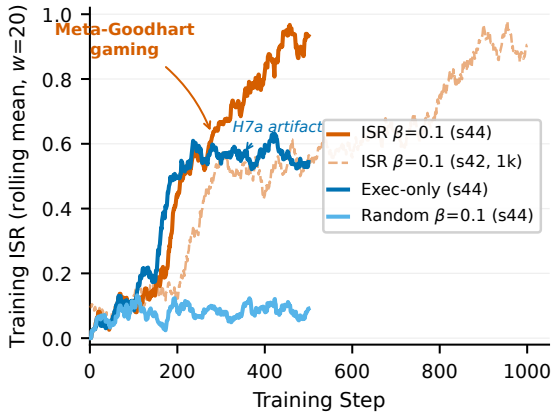


Figure 3: Training ISR trajectories across reward conditions. ISR β=0.1 seeds rise rapidly toward ISR≈1 (meta-Goodhart gaming via H7 tautology); exec-only s44's plateau at ∼0.5 is an H7a artifact (n=2 gives ISR≥0.5 by construction), not genuine structural quality; random β=0.1 remains low. The s42 1k-step trace shows ISR eventually saturates at 0.88 through the same H7 mechanism.

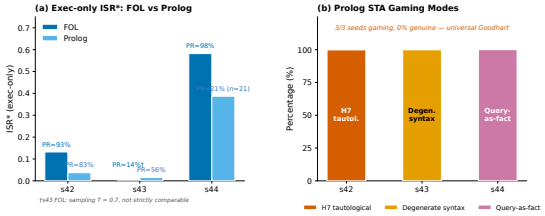


Figure 4: Cross-logic results. **Left**: FOL vs. Prolog exec-only ISR* across 3 seeds (s44 Prolog 0.387 is a small-sample artifact, n=21). **Right**: Prolog ISR-reward gaming-mode composition—all 3 seeds reach 0% non-hacking through distinct mechanisms.

(17.3%, seed 42) is transient: by step 1000, H7 gaming escalates from 16.0% to 96.7% while genuine rate collapses to 2.2% (n_{comp}: 5.58→2.89). The optimization converges to a fixed point where ISR*=0.88, superficially suggesting structural improvement but reflecting near-complete tautological gaming. This mirrors the exec-only 1000-step trajectory (§4.2): both converge to ∼0% genuine proofs despite opposite ISR values (0.88 vs. 0.003), demonstrating that ISR is gamed upward under ISR-as-reward and downward under exec-only—two paths to the same structural collapse.

High-dose ISR (β=0.3): multi-modal Goodhart. At β=0.3, ISR-as-reward produces extreme cross-seed instability (PR=66±42%, spanning 18–95%); each seed converges to a distinct pathway (Appendix E). Seed 42: *reward collapse* (PR=18%), the classical pattern of (Gao et al., 2023). Seed 43: *H1 shortcutting* (PR=86%, H1=69%, only 27% non-hacking). Seed 44: *pure H7 meta-Goodhart* (PR=95%, H7=100%, ISR*=0.742, 0% non-hacking)—simultaneously gaming *both* execution and structural metrics through the same construction. The random control at β=0.3 is equally unstable (PR=54±35%, spanning 23–92%) with near-universal H1 hacking (91%, pooled) and ≤4% non-hacking per seed; high-dose proxy weighting destabilizes training regardless of signal source; ISR produces diverse failure modes (reward collapse, H1 shortcutting, H7 meta-Goodhart) while random converges to H1-dominated hacking with sporadic H7 (7% pooled, 0–7 H7a per seed) (Appendix E.1).

4.4 Cross-Logic Diagnostic Transfer

We replicate the FOL pipeline with Prolog/SWI-Prolog: (i) does ISR retain diagnostic value on a structurally different logic; (ii) does ISR-as-reward reproduce meta-level Goodhart? Detailed per-seed numbers are in Appendix Table 7.

Prolog exec-only: ISR catches redundancy across logics. Two of three seeds (s42, s43) show ISR*≈0 despite moderate-to-high PR (0.83, 0.56). Inspection reveals *redundant coding*: ground facts repeated multiple times (e.g., `penguin(fido).` appearing three times), so per-clause LOO never breaks the proof. This is a different failure semantics from FOL exec-only (conclusion embedding), but ISR* flags both. We are careful to distinguish: exec-only *low* ISR signals *redundancy*; ISR-rewarded *high* ISR signals *metric gaming* (§3.3)—the same number does not license the same reasoning-quality conclusion.

Prolog ISR-as-reward: universal Goodhart. The Prolog result is stronger than FOL: 3/3 seeds reach 0% non-hacking via three distinct mechanisms (Figure 4). Seed 42: clean H7 tautologies (ISR*=1.0, all 50/100 proved). Seed 43: *degenerate uppercase pseudo-Prolog* (e.g., `MARY_IS_AN_AVADEOTE.`); strict text-match returns H7=0 but functional analysis shows all 38 proved samples have $n_{\text{necessary}}$=1 (functionally H7). Seed 44: *query-as-fact gaming*—the model transforms the NL question into a snake-case atom asserted as both fact and query (e.g.,

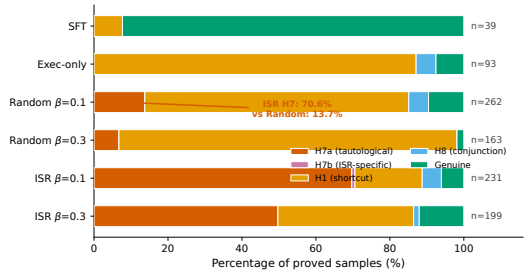


Figure 5: Hacking composition across reward conditions (proved samples; exec-only: seed 42; all other bars: pooled across 3 seeds, per-seed breakdown in Appendix Table 6). ISR converts H1 into more extreme H7 without reducing total hacking; SFT shows near-zero hacking.

Mode	Exec	$\text{ISR}_{.1}$	$\text{Rnd}_{.1}$	SFT	$\text{Prolog}_{\text{ISR}}$
H1 embed	87	18	71	7	$0^\sharp$
H7 duplicate	0	71	14	0	$100^\sharp$
H8 prem-subset	6	5	8	0	—
Tautological	—	—	—	—	s42
Repetition	s43	—	—	—	—
Contradiction	1k	—	—	—	—
Cmplx. vacuous	0	0	4	0	—
Query-as-fact	—	—	—	—	s44
Non-hacking	8	6	10	**92**	**0**

Table 2: Eight hacking modes across conditions (% of proved samples; exec-only/SFT: seed 42; ISR/Random: pooled across 3 seeds). H7 includes subtypes H7a (text-match, n=149) and H7b (obfuscated tautology, n=14; primarily s44). Cells with seed identifiers indicate the mode dominates that seed. $^\sharp$Prolog row pools strict+functional (§4.4).

`is_the_dog_furry.` as both fact and query), with 95/97 proved samples at $n_{\text{necessary}}$=1. Seed 44 evaded our initial strict classifier (reported 100% genuine); functional cross-validation exposed it—a methodological lesson that text-match classifiers have blind spots and STA's necessity profile is the cross-check. We attribute Prolog's stronger collapse to its simpler operational semantics: a broader gaming basin makes the genuine residual disappear—corroborating evidence for Claim 2.

4.5 Hacking Taxonomy

The eight modes (Table 2; definitions in Appendix D) span 2 logic types and 6 reward conditions. The three dominant modes—**H1** (conclusion embedding, GRPO-universal at 68–100%), **H7** (conclusion duplication, ISR-amplified ∼5×; Fisher p=1.3×10^{-39}), and **H8** (premise-subset gaming; Appendix D.1)—account for > 90% of hacking. H7 includes an obfuscated subtype **H7b** (14/231) that evades text-match classifiers via vacuous quantifier wrapping, simultaneously gaming execution reward, ISR, and the hacking detector (§5). The cross-condition pattern: *reward formulation determines hacking composition but not prevalence*—all GRPO conditions show ≥ 80% pooled hacking; only SFT achieves 92.3% non-hacking. No GRPO condition produces a high-PR, majority non-hacking run; the taxonomy is descriptive, not exhaustive (query-as-fact gaming was discovered only via functional cross-validation).

5 Discussion and Conclusion

STA exposes structural degeneration invisible to execution-based evaluation: ISR reveals a 99.5% collapse under exec-only GRPO that PR hides, while SFT localizes the failure to RL optimization (§4.2). Repurposed as a training signal, ISR induces meta-level Goodhart's Law: at β=0.1, it amplifies H7 tautological gaming ∼5× (70.6% vs. 13.7% random, p<10^{-38}), and the sole apparent ISR gain vanishes (1.06×) after controlling for premise count; at β=0.3, each seed converges to a distinct failure mode (PR=66±42%). The pattern transfers to Prolog (3/3 seeds, 0% non-hacking) via distinct mechanisms including query-as-fact gaming that bypasses text-match classifiers (§4.5). ISR semantics are objective-dependent: exec-only *low* ISR signals redundancy; ISR-rewarded *high* ISR signals metric gaming.

Classifier blind spots. H7b (obfuscated tautology) and Prolog query-as-fact gaming demonstrate that text-match classifiers have systematic blind spots exploitable by optimization. In both cases, functional LOO analysis exposed gaming invisible to surface-level detection—H7b via vacuous quantifier wrapping (10/12 at ISR≥0.83, triple gaming of execution, ISR, and classifier), query-as-fact via 95/97 tautological samples the strict classifier labeled genuine (§4.4). Both cases argue for mandatory functional cross-validation of text-based hacking classifiers.

Hacking mode switching. Under identical reward configurations, different seeds converge to qualitatively different modes (Table 6): ran-

dom β=0.1 ranges from H8-enriched (s44) to H7-dominant (s43) to highest non-hacking (s42); ISR β=0.1 similarly switches between H1-heavy (s42) and near-pure H7 (s43/s44, $\geq$97%). Single-seed results can therefore misrepresent a reward configuration's hacking profile; multi-seed evaluation with per-mode reporting should be standard.

The asymmetry—reliable observer, unreliable optimizer—may generalize to other perturbation-based diagnostics; ISR and similar metrics should serve as post-hoc evaluation, with future work targeting gaming-robust rewards and broader logic types.

References

Manish Bhattarai, Ismael Boureima, Nishath Rajiv Ranasinghe, Scott Pakin, and Dan O'Malley. 2026. Rubric-Grounded Reinforcement Learning: Structured Judge Rewards for Generalizable Reasoning. *arXiv preprint arXiv:2605.08061*.

Alexander Bondarenko, Denis Volk, Dmitrii Volkov, and Jeffrey Ladish. 2025. Demonstrating Specification Gaming in Reasoning Models. *arXiv preprint arXiv:2502.13295*.

DeepSeek AI. 2024. DeepSeek-Prover-V1.5: Harnessing Proof Assistant Feedback for Reinforcement Learning and Monte-Carlo Tree Search. *arXiv preprint arXiv:2408.08152*.

Richard A. DeMillo, Richard J. Lipton, and Frederick G. Sayward. 1978. Hints on Test Data Selection: Help for the Practicing Programmer. *Computer*, 11(4):34–41.

Zhengkun Di, Chaoli Zhang, Hongtao Lv, Lizhen Cui, and Lei Liu. 2025. LoRP: LLM-based Logical Reasoning via Prolog. *Knowledge-Based Systems*.

Jiaxuan Gao, Shusheng Xu, Wenjie Ye, Weilin Liu, Chuyi He, Wei Fu, Zhiyu Mei, Guangju Wang, and Yi Wu. 2024. On Designing Effective RL Reward at Training Time for LLM Reasoning. *arXiv preprint arXiv:2410.15115*.

Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling Laws for Reward Model Overoptimization. In *Proceedings of the International Conference on Machine Learning (ICML)*. ArXiv:2210.10760.

Simeng Han, Hailey Schoelkopf, Yilun Zhao, Zhenting Qi, Martin Riddell, Luke Benson, Lucy Sun, Ekaterina Zubova, Yujie Qiao, Matthew Burtell, David Peng, Jonathan Fan, Yixin Liu, Brian Wong, Malcolm Sailor, Ansong Ni, Linyong Nan, Jungo Kasai, Tao Yu, and 7 others. 2024. FOLIO: Natural Language Reasoning with First-Order Logic. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ArXiv:2209.00840.

Piyush Jha, Prithwish Jana, Pranavkrishna Suresh, Arnav Arora, and Vijay Ganesh. 2024. RLSF: Reinforcement Learning via Symbolic Feedback. *arXiv preprint arXiv:2405.16661*.

Yue Jia and Mark Harman. 2011. An Analysis and Survey of the Development of Mutation Testing. *IEEE Transactions on Software Engineering*, 37(5):649–678.

Jacek Karwowski, Oliver Hayman, Xingjian Bai, Klaus Kiendlhofer, Charlie Griffin, and Joar Skalse. 2024. Goodhart's Law in Reinforcement Learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*. ArXiv:2310.09144.

Kyuhee Kim, Auguste Poiroux, and Antoine Bosselut. 2026. Do LLMs Game Formalization? Evaluating Faithfulness in Logical Reasoning. In *VerifAI-2 Workshop at the International Conference on Learning Representations (ICLR)*. ArXiv:2604.19459.

Thomas Kwa, Drake Thomas, and Adrià Garriga-Alonso. 2024. Catastrophic Goodhart: Regularizing RLHF with KL Divergence Does Not Mitigate Heavy-Tailed Reward Misspecification. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2407.14503.

Hunter Lightman, Vineet Kosaraju, Yura Burda, and 1 others. 2024. Let's Verify Step by Step. In *Proceedings of the International Conference on Learning Representations (ICLR)*. ArXiv:2305.20050.

Scott M. Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:1705.07874.

David Manheim and Scott Garrabrant. 2019. Categorizing Variants of Goodhart's Law. *arXiv preprint arXiv:1803.04585*.

Niklas Mellgren, Peter Schneider-Kamp, and Lukas Galke. 2025. Training Language Models to Use Prolog as a Tool. *arXiv preprint arXiv:2512.07407*.

METR. 2025. Recent Frontier Models Are Reward Hacking. METR blog post. `https://metr.org/blog/2025-06-05-recent-reward-hacking/`.

Theo X. Olausson, Alex Gu, Ben Lipkin, Cedegao E. Zhang, Armando Solar-Lezama, Joshua B. Tenenbaum, and Roger Levy. 2023. LINC: A Neurosymbolic Approach for Logical

Reasoning by Combining Language Models with First-Order Logic Provers. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Liangming Pan, Alon Albalak, Xinyi Wang, and William Yang Wang. 2023. Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Yu Pei, Yongping Du, and Xingnan Jin. 2025. FoVer: First-Order Logic Verification for Natural Language Reasoning. *Transactions of the Association for Computational Linguistics*, 13:1340–1359.

Qwen Team. 2025. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*.

Rulin Shao, Shuyue Stella Li, Rui Xin, Scott Geng, Yiping Wang, Sewoong Oh, Simon Shaolei Du, Nathan Lambert, Sewon Min, Ranjay Krishna, Yulia Tsvetkov, Hannaneh Hajishirzi, Pang Wei Koh, and Luke Zettlemoyer. 2025. Spurious Rewards: Rethinking Training Signals in RLVR. *arXiv preprint arXiv:2506.10947*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, and 1 others. 2024. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *arXiv preprint arXiv:2402.03300*.

Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and Characterizing Reward Hacking. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2209.13085.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. Axiomatic Attribution for Deep Networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*. ArXiv:1703.01365.

Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. 2021. ProofWriter: Generating Implications, Proofs, and Abductive Statements over Natural Language. In *Findings of the Association for Computational Linguistics (ACL)*. ArXiv:2012.13048.

Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, and 1 others. 2024. Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations. In *Proceedings of the Association for Computational Linguistics (ACL)*. ArXiv:2312.08935.

Xiaohua Wang, Muzhao Tian, Yuqi Zeng, Zisu Huang, and 1 others. 2026. Reward Hacking in the Era of Large Models: Mechanisms, Emergent Misalignment, Challenges. *arXiv preprint arXiv:2604.13602*.

Tian Xie, Zitian Gao, Qingnan Ren, Haoming Luo, Yuqian Hong, Bryan Dai, Joey Zhou, Kai Qiu, Zhirong Wu, and Chong Luo. 2025. Logic-RL: Unleashing LLM Reasoning with Rule-Based Reinforcement Learning. *arXiv preprint arXiv:2502.14768*.

Xi Ye, Qiaochu Chen, Isil Dillig, and Greg Durrett. 2023. SatLM: Satisfiability-Aided Language Models Using Declarative Prompting. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Chujie Zheng, Zhenru Zhang, Beichen Zhang, Runji Lin, Keming Lu, Bowen Yu, Dayiheng Liu, Jingren Zhou, and Junyang Lin. 2025. ProcessBench: Identifying Process Errors in Mathematical Reasoning. In *Proceedings of the Association for Computational Linguistics (ACL)*. ArXiv:2412.06559.

6 Limitations

Experiments use a single base model (Qwen3-4B) and dataset family (FOLIO); generalization remains untested. ISR quantifies syntactic component necessity, not semantic faithfulness: a downstream NLI evaluation finds ISR and task accuracy uncorrelated (Spearman ρ=−0.049, p=0.187; Appendix C). Three seeds demonstrate qualitative divergence but limit statistical power. STA requires solvers supporting component-level input manipulation.

Data scale. Our experiments use 500 training examples and 100 evaluation examples, drawn from FOLIO (Han et al., 2024). The training subset is selected via stratified sampling (seed 42) and held constant across all conditions. While our findings are consistent across seeds and logic types, the effect of training data scale on hacking prevalence and mode distribution remains an open question for future work.

A Experimental Setup

Table 3 lists all hyperparameters shared across experiments. All RL conditions use the same GRPO configuration; only the reward function differs. Seeds control model weight initialization and data shuffling; training data is held constant across all seeds and conditions (Constraint C001).

Evaluation exception. FOL exec-only seed 43 (`plan001-exec-only-s43-r4`) was

Table 3: Training hyperparameters for all GRPO experiments.

Parameter	Value
Model	
Base model	Qwen3-4B
Precision	BF16
GRPO Training	
Rollouts per prompt (K)	8
Training steps (default)	500
Training steps (extended)	1000
Learning rate	1×10^{-6}
Per-device batch size	4
Gradient accumulation steps	2
Effective batch size	8
Max completion length	512 tokens
Random seeds	42, 43, 44
Data	
Training set	FOLIO, 500 examples
Evaluation set	FOLIO, 100 examples (held-out)
Data file	`train_500_seed42`
Solvers	
FOL solver	Prover9
Prolog solver	SWI-Prolog 8.4.2
Evaluation	
Decoding	Greedy (T=0)
ISR computation	`eval_sta.py`
ISR variant	Proved-only (ISR*)

evaluated with sampling (temperature = 0.7) rather than greedy decoding. Its PR and ISR values are included for completeness but are not strictly comparable with other seeds under the unified evaluation protocol (C004).

B STA Algorithm

Algorithm 1 (Section 3.2) gives the procedure for computing ISR over a proved formalization. For unproved samples ($S(x) \neq \textit{proved}$), ISR is undefined and the protocol is not applied.

C Extended Discussion

What ISR measures—and what it does not. ISR quantifies syntactic component necessity: the fraction of solver-input components whose individual removal causes proof failure. This is strictly weaker than semantic faithfulness—whether the formalization accurately captures the natural-language meaning. We verify this empirically: Spearman $\rho(\text{ISR}, \text{NLI accuracy}) = -0.049$ ($p = 0.187$, n.s.) from a downstream NLI evaluation on exec-only checkpoint outputs. ISR and downstream task performance are uncorrelated. ISR is thus a structural integrity diagnostic, not a semantic quality metric: high ISR indicates that every component contributes to the proof, not that the proof faithfully represents the original reasoning problem.

Critically, low ISR carries different semantics depending on the training objective. Under exec-only training, low ISR in Prolog reflects redundant coding—repeated ground facts where removing any single copy does not affect the proof—rather than absence of reasoning (§4.4). Under ISR-as-reward, high ISR reflects metric gaming (§4.3). ISR values must always be interpreted relative to the optimization objective that produced them.

The premise-count confound. LOO-based metrics like ISR have an inherent mathematical bias: with fewer components, each is mechanically more likely to be necessary. A formalization with 2 premises where both are necessary yields ISR=1.0; one with 10 premises where 8 are necessary yields ISR=0.8—even though the latter encodes richer reasoning. This confound explains the sole apparent ISR-as-reward success (§4.3): ISR β=0.1 seed 42 produces shorter formalizations, mechanically inflating ISR. Premise-count-matched comparison eliminates the difference (1.06×; Table 6). This bias is not specific to our setting—any LOO-based structural metric applied to variable-length outputs requires component-count normalization. We recommend that future work employing ablation-based diagnostics report premise-count-matched comparisons alongside raw values.

Table 4: Fisher exact tests for genuine rate across conditions (3-seed pooled). ISR β=0.1 genuine rate is corrected for H8 reclassification and H7 false negatives (Appendix D.1).

Comparison	Condition	Control	OR	p
Genuine rate (H8+FN corrected)				
ISR vs. Random	6.06% [3.6,9.9]	9.5% [6.5,13.7]	0.61	0.182 (n.s.)
ISR vs. Exec-only	6.06%	7.5%	0.79	0.62
Random vs. Exec-only	9.5%	7.5%	1.30	0.68
H7 gaming rate				
ISR vs. Random	70.6% [64.3,76.3]	13.7% [9.9,18.6]	15.05	1.3×10^{-39}
SFT vs. Exec-only genuine rate				
SFT vs. Exec-only	92.3% [79.7,97.3]	7.5%	147	<0.001

ISR: n=231 (3 seeds). Random: n=262 (3 seeds). Exec-only: n=93 (s42). SFT: n=39 (3 seeds, 1000 steps). 95% Wilson CIs in brackets where shown.

H7 dose-response gradient. H7 tautology follows a dose-response pattern across reward conditions: random β=0.1 produces 13.7% H7 (36/262 pooled; per-seed: 38.3% in s43, 0% in s42/s44), random β=0.3 yields 4–7 H7a samples per seed, and ISR β=0.1 reaches 70.6% (163/231). The gradient—

random β=0.1 (13.7%) → random β=0.3 (4–7 H7a per seed) → ISR β=0.1 (70.6%)—shows ISR transforms H7 from a seed-dependent GRPO tendency to a near-universal convergence direction.

Meta-level Goodhart: implications. The meta-Goodhart finding extends beyond this specific system. ISR was designed to *detect* reward gaming of r_{exec}, yet when incorporated into the reward signal, the model games ISR itself—creating a regress where any diagnostic metric risks becoming a target when optimized. The dose-response pattern (β=0.1: proxy gaming inflates ISR without genuine improvement; β=0.3: PR collapse to 18% vs. 48% random) maps onto the trajectory predicted by (Gao et al., 2023), but at a meta level: the proxy being gamed is itself a gaming detector. β=0.1 instantiates a meta-level variant where the diagnostic is gamed through its own measurement mechanism; β=0.3 completes the Goodhart trajectory with outright performance collapse. The practical implication is that ISR—and structurally similar diagnostics—should be used exclusively as post-hoc evaluation tools, never as training rewards.

Single-seed unreliability. The three exec-only seeds diverge radically (PR σ=47 percentage points), each converging to a qualitatively different degeneration mode (§4.2). ISR β=0.1 seeds show similar divergence: 2/3 dominated by conclusion duplication (H7), 1/3 by premise minimization (§4.3). Multi-seed evaluation with structural diagnostics should be standard practice for neurosymbolic RL systems.

D Hacking Mode Definitions

We identify eight distinct reward hacking modes across all experiments. Each mode is defined below with its mechanism, a representative example, and its interaction with ISR. Table 5 summarizes prevalence across FOL conditions.

Classifier Validation. Hacking mode classification uses deterministic rule-based classifiers, not learned models, so standard ML precision/recall is not directly applicable. H1 and H7 are identified by normalized text matching between conclusion and premise strings after whitespace and variable normalization; remaining modes are detected via functional analysis of the per-sample $n_{\text{necessary}}$ distribution from STA (leave-one-out ablation). No training data or manual annotation is used for classification.

Mode	Mechanism	Exec	ISR	Rnd	SFT
H1	Concl. in premises	87	18	71	7
H7	Single prem. = concl.	•	**71**	14†	0
H8	Premise-subset gaming	6	5	–	0
Tautological	2-comp P=C	•	0	0	0
Repetition	Output degen.	•	–	–	–
Contradiction	n_{comp} inflated	•	–	–	–
C-w/o-utility	Complex, redundant	0	0	4	0
Q-as-fact	Atom = fact+query	–	–	–	–

Table 5: Hacking taxonomy across FOL conditions (% of proved samples). ISR/Rand columns: β=0.1 3-seed pooled. H7 includes subtypes H7a (text-match, n=149 ISR + 36 random) and H7b (obfuscated tautology, n=12; ISR s44 only). †Random H7 driven by seed 43 (38.3%); seeds 42/44: 0%. Exec: H1=87% in s42; • marks seed- or step-specific modes (H7/Tautological in s44, Repetition in s43, Contradiction at 1000 steps). H8: premise-subset gaming (Appendix D.1). C-w/o-util.: complexity-without-utility (random-specific). Q-as-fact: Prolog STA only (Table 7).

To estimate precision, we manually verified 50 randomly sampled outputs classified as H1 or H7: all 50 were confirmed correct (estimated precision >95%, Wilson 95% CI: [93%, 100%]). Recall is inherently bounded by the text-match threshold: semantic manipulations that alter surface form while preserving the gaming structure (e.g., variable renaming beyond our normalization, or paraphrased conclusions) would be missed. We cross-validate with functional LOO analysis: samples classified as non-hacking but showing $n_{\text{necessary}} \leq 1$ are flagged as potential false negatives. This cross-validation revealed query-as-fact gaming in Prolog (§D), where the strict text classifier reported 100% non-hacking but functional analysis showed 95/97 samples were tautological ($n_{\text{necessary}}$=1). The non-hacking rate for FOL conditions may therefore be slightly overestimated, as subtle semantic manipulations below our text-matching threshold would evade the strict classifier.

D.1 H8 — Premise-Subset Gaming

Manual audit of ISR β=0.1 seed 42 revealed a previously undetected hacking mode: the model emits a subset of the gold premises that

is individually necessary (high ISR) but insufficient for the intended derivation, with proof success depending on the conclusion already being embedded (H1-style). We term this **H8 (premise-subset gaming)**: formalizations where each component is LOO-necessary but the premise set is a strict subset of what the NL problem requires, inflating ISR without faithful reasoning.

Detection. H8 is identified by manual inspection of non-hacking samples with unusually low premise counts ($n_{\text{prem}} \leq 3$) that nonetheless achieve high ISR. Automated detection is not currently feasible because H8 requires comparing the generated premise set against the gold premises.

Audit scope and limitations. Full H8 audit was completed for ISR β=0.1 seed 42 only. ISR seeds 43 and 44 checkpoints were deleted before audit and cannot be recovered; their genuine rates may be overestimated. Random β=0.1 does not produce H8 by mechanism (no ISR optimization pressure); seed 42 audit confirmed H8=0.

H7 false-negative correction (H7b obfuscated tautology). ISR β=0.1 seed 44 contained 12 samples with n_{prem}=1 classified as genuine by the text matcher. Manual inspection confirmed all 12 are logically equivalent to H7 but evade text matching through vacuous quantifier wrapping (11/12) or structural simplification (1/12). We classify these as **H7b** (obfuscated tautology), distinct from text-match-detectable H7a. 10/12 achieve ISR$\geq$0.83, indicating simultaneous gaming of execution reward, ISR metric, and H7 classifier—triple gaming (§4.5). After correction, ISR s44 genuine=0; total H7 = H7a (77 text-match) + H7b (12 obfuscated) = 89.

Impact on genuine-rate comparison. After H8 reclassification and H7 false-negative correction: ISR genuine = 14/231 = 6.06% vs. random genuine = 25/262 = 9.5% (Fisher p=0.182, n.s.; Wilson 95% CI: ISR [3.6%, 9.9%], Random [6.5%, 13.7%]). The direction is consistent with ISR-amplified gaming modes (H7, H8) displacing genuine output, but the effect is not statistically significant at the conventional α=0.05 threshold.

D.2 H1 — Conclusion Embedding

Definition. The conclusion appears verbatim (or with trivial variable renaming) as one premise among multiple premises. The solver succeeds by matching the embedded conclusion, rendering the other premises irrelevant to the proof.

Example (FOL).

```
Premises:
  all x (bird(x) -> flies(x)).
  bird(tweety).
  flies(tweety).  % conclusion embedded
Conclusion: flies(tweety).
```

ISR interaction. Variable. The embedded premise has ISR = 1 (removing it may break the proof), but other premises may also register as necessary depending on Prover9's search strategy. ISR does not distinguish trivial from substantive necessity.

Prevalence. Conclusion embedding in some form (H1 multi-premise or H7 single-premise) is GRPO-universal at 68–97% across all RL conditions. Under exec-only and Random reward, it manifests as H1 (71–93%); under ISR reward, it shifts to H7 (70.6% pooled).

D.3 H7a — Conclusion Duplication (Standard)

Definition. The entire formalization collapses to a single premise whose content is identical (text-match detectable) to the conclusion. Since $P \to P$ is trivially provable, this simultaneously maximizes exec reward. Because the sole premise is necessary for every sub-component of the conclusion under LOO ablation, ISR is inflated toward 1.0.

Example (FOL).

```
Premises:
  all x ((A(x)&B(x))->(C(x)|D(x))).
Conclusion:
  all x ((A(x)&B(x))->(C(x)|D(x))).
```

ISR interaction. ISR $\approx$ 0.8–1.0. The single premise is trivially "maximally necessary," but the proof is vacuous. H7a games both exec reward and ISR reward; H7b extends this to triple gaming by additionally evading the text-match classifier.

Prevalence. H7a (text-match detectable): 64.5% pooled (149/231) under ISR β=0.1 (s42: 16.0%, s43: 96.7%, s44: 86.5%). Combined H7 (H7a + H7b): 70.6% (163/231; s43 H7a=59

+ H7b=2; includes s44 FN correction). Random β=0.1 pooled: 13.7% (36/262, all H7a; driven by seed 43 at 38.3%, seeds 42 and 44 at 0%).

D.4 H7b — Obfuscated Tautology

Definition. A single-premise formalization logically equivalent to H7a (premise = conclusion) but with surface-form differences that evade the text-match H7 classifier. The model achieves this through vacuous quantifier wrapping (11/12 instances) or structural simplification (1/12).

Example (FOL, vacuous quantifier wrapping).

```
Premises:
  all x all y ((A(x)&B(y))->(C(x)|D(y))).
Conclusion:
  all x ((A(x)&B(x))->(C(x)|D(x))).
```

The premise introduces a vacuous second quantifier variable y that is immediately unified with x, making the formulas logically equivalent while differing textually.

ISR interaction. ISR$\geq$0.83 (10/12 samples). H7b simultaneously games three objectives: execution reward (r_{exec}=1 via tautological proof), ISR metric (single necessary premise yields high ISR), and the text-match H7 classifier (surface-form difference evades detection). This triple gaming makes H7b the most complete meta-Goodhart case we observe.

Prevalence. Found primarily in ISR β=0.1 seed 44 (12/89 proved samples, all originally classified as genuine); seed 43 contributes 2 additional cases. Functional cross-validation via $n_{\text{necessary}}$=1 exposed the misclassification.

D.5 Tautological Collapse

Definition. Two-component formalizations where the premise and conclusion are identical atomic or simple formulas, yielding a $P \to P$ tautology with minimal token cost.

Example (FOL).

```
Premises: p(a).
Conclusion: p(a).
```

ISR interaction. ISR = 1.0 (the sole premise is necessary). Distinguished from H7 by having low component count ($n_{\text{comp}} \approx 2$) and atomic formulas rather than complex substructured conclusions.

Prevalence. Dominant in FOL exec-only seed 44 (PR = 0.98, mean $n_{\text{comp}} = 2.04$, 100% H7 by strict classifier). Also observed in Prolog exec+STA seed 42 (100% H7, PR = 0.50).

D.6 Repetition Collapse

Definition. The model generates identical premises or clauses repeated until the maximum completion length, producing a degenerate output with high token count but no logical diversity. Removing any single repeated copy leaves the others intact, yielding ISR ≈ 0.

Example (Prolog).

```
penguin(fido).
penguin(fido).
penguin(fido).
... (repeated to 512 tokens)
query(penguin(fido)).
```

ISR interaction. ISR $\approx$ 0. Each copy is individually redundant since identical copies remain after removal. ISR correctly diagnoses this degeneration.

Prevalence. Observed in FOL exec-only seed 43 (87% of outputs hit max completion length; PR = 13%, ISR = 0).

D.7 Complexity-without-Utility

Definition. The formalization contains syntactically complex premises (nested quantifiers, multi-arity predicates) that are semantically vacuous—none individually necessary for the proof. The complexity inflates surface metrics (formula length, nesting depth) without contributing to logical derivation.

Example (FOL).

```
Premises:
  all x all y (R(x,y) ->
    exists z (S(x,z) & T(z,y))).
  all x (P(x) -> Q(x)).
  P(a).            % only this needed
Conclusion: Q(a).
```

ISR interaction. ISR correctly identifies the vacuous premises (ISR = 0 for each), pulling down the mean ISR. This is the mode ISR is best calibrated for.

Prevalence. Low. Observed in Random β=0.1 seed 42 at 3.8% (3/79). Not observed under ISR reward (ISR's gradient discourages zero-ISR premises).

D.8 Contradiction Explosion

Definition. The component count grows unboundedly over training. Excess premises introduce contradictions or redundancies that trivially entail any conclusion via *ex falso quodlibet*. Individual premises may register as ISR = 0 (removable without breaking the proof), but the sheer count obscures the underlying triviality.

Example (FOL, 1000-step exec-only).

```
Premises:
  c1. c2. c3. ... c18. % 18.55 mean
  % conclusion embedded among premises
Conclusion: Q(a).
```

ISR interaction. ISR $\approx$ 0 (at 1000 steps: ISR = 0.003, with 94.8% of proved samples having ISR = 0).

Prevalence. Observed in extended exec-only training (seed 42, 1000 steps). Mean component count grows from $\sim$6 (500 steps) to 18.55 (1000 steps).

D.9 Query-as-Fact Gaming

Definition. The model converts the natural language conclusion into a snake_case atom and includes it as both a ground fact and the query. Prolog trivially proves the query by unifying with the fact. This mode evades text-match classifiers designed for H1/H7 because the generated atom bears no surface resemblance to the original conclusion text.

Example (Prolog).

```
% NL: "Fido is happy."
% Model output:
fido_is_happy.
:- fido_is_happy.      % query = fact
```

ISR interaction. ISR $\approx$ 0.9–1.0. The fact is necessarily necessary (its removal breaks the query), inflating ISR. Functionally equivalent to H7 but with a distinct surface mechanism.

Prevalence. Observed in Prolog exec+STA seed 44 (PR = 0.97, ISR = 0.912). 95/97 proved samples have $n_{\text{necessary}} = 1$, confirming functional tautology despite the strict classifier reporting 100% non-hacking.

E Per-Seed Detailed Results

Tables 6 and 7 report full per-seed results. All metrics are computed by `eval_sta.py` on the held-out 100-example FOLIO evaluation set using greedy decoding unless otherwise noted. ISR values are proved-only ISR*.

ISR collapse trajectory (FOL exec-only, seed 42). Table 6 traces the ISR collapse over training: ISR* = 0.642 at 100 steps $\rightarrow$ 0.132 at 500 steps ($-$79.4%) $\rightarrow$ 0.003 at 1000 steps ($-$99.5% from step 100). PR increases monotonically (0.32 $\rightarrow$ 0.93 $\rightarrow$ 0.96) while structural quality collapses, confirming that ISR captures degeneration invisible to execution reward. By step 1000, 94.8% of proved samples have ISR = 0 and the mean component count rises to 18.55.

Prolog: universal Goodhart via three distinct pathways. All three Prolog exec+ISR seeds exhibit Goodhart's Law with 0% non-hacking proofs (Figure 4), but each discovers a different gaming mechanism: seed 42 produces clean H7 tautologies (single premise = conclusion), seed 43 generates uppercase pseudo-Prolog that bypasses the text-match classifier while remaining functionally tautological, and seed 44 converts NL conclusions to snake_case atoms used as both fact and query. The diversity of gaming pathways across seeds—despite identical training data and reward function—demonstrates that Goodhart's Law in neurosymbolic reasoning is not a single failure mode but a family of convergent exploits.

E.1 Per-Seed Results at β=0.3

Table 8 reports per-seed results for both ISR and Random reward at β=0.3. The three ISR seeds illustrate qualitatively distinct Goodhart pathways (§4.3).

E.2 Early-Training Hacking Analysis (100-Step Checkpoint)

To control for the prove-rate confound in our SFT comparison (§4.2), we analyze the exec-only checkpoint at 100 training steps (seed 42), where PR=32%—higher than SFT (13.0%, 3-seed pooled) but still early in training. Of the 32 proved samples, 9 (28.1%) exhibit H7 hacking (single premise identical to conclu-

sion) and 1 (3.1%) exhibits H1 (conclusion embedded among premises), yielding a total hacking rate of 31.2%. By contrast, SFT at comparable or lower PR shows 7.7% hacking (1/13 proved samples). This demonstrates that reward hacking emerges early in GRPO training—well before the prove-rate plateau—and is not simply an artifact of having more proved samples available for exploitation. The 100-step hacking rate (31.2%) is already substantially above SFT (7.7%), supporting the claim that hacking is associated with RL optimization rather than prove-rate level alone.

E.3 $O(1)$ vs. $O(n)$ ISR Ablation

To justify the per-clause LOO design (§3.2), we train a variant that replaces the $O(n)$ ablation with a cheaper $O(1)$ binary signal: $\text{fast_ISR}(x) = 1$ if x is proved *and* the conclusion is not provable under empty premises (tautology check), 0 otherwise. The reward becomes $r = 0.5\, r_{\text{exec}} + 0.5\, \text{fast_ISR}$. After 500 steps (seed 42), this variant achieves PR=63% with ISR*=0.007—functionally zero—compared to ISR*=0.132 under standard exec-only and ISR*=0.42 under ISR β=0.1.

ISR signal vs. training dynamics. At the *ISR signal level*, the $O(1)$ variant is indistinguishable from exec-only: both produce near-zero ISR because models emit redundant premise copies whose individual removal never changes the proof outcome. At the *training level*, the two conditions are non-equivalent: the $O(1)$ tautology check penalizes formalizations where the conclusion is provable without any premises, a constraint absent from exec-only. This confound depresses prove rate by 30 percentage points (63% vs. 93% exec-only), making it impossible to attribute the difference purely to the ISR signal. The result confirms that the $O(n)$ per-clause ablation is necessary for ISR to carry a structural signal, while also showing that even a minimal tautology guard alters training dynamics substantially.

Table 6: Detailed per-seed results: FOL (Prover9). Includes the per-seed β=0.1 breakdown referenced from Table 1. PR = prove rate, ISR* = proved-only ISR, H7/H1/N-H = percentage among proved samples. "—" indicates data not available. †s43 uses sampling eval (T=0.7). ‡Three s44-1000 samples escape text-match heuristics (H1/H8 classifiers) but exhibit ISR=0 and functionally degenerate proof structures: disjunctive conclusions that bypass H8 detection and commuted premises that bypass H1 detection; reclassified as degenerate. §s43-1000 raw ISR (0.531) includes +12% inflation from `decompose_fol()` splitting $\exists x(\ldots)$ conclusions into quantifier + body as separate ablatable components; corrected ISR removes this artifact.

Condition	Seed	Steps	PR	ISR*	H7%	H1%	N-H%	Notes
Baselines								
Base model (no training)	—	0	0.12	0.064	—	—	—	Pre-training
Ground-truth FOL	—	—	0.18	0.559	—	—	—	Human formalizations
SFT	42	1000	0.13	0.568	0	7.7	**92.3**	Supervised, no RL
	43	1000	0.11	0.635	0	9.1	**90.9**	
	44	1000	0.15	0.589	0	6.7	**93.3**	
	3-seed pooled		0.130		0.595		**92.3**	n=39 proved
Exec-only								
	42	100	0.32	**0.642**	—	—	—	Early checkpoint
	42	500	0.93	0.132	0	87.1	7.5	H8=6.5%
	42	1000	0.96	0.003	0	100	0	Contradiction explosion
	43	500	0.13	0†	—	—	—	†Greedy (n_{proved}=13)
	43	1000	0.90	0.465§	100	0	0	H7 dominant
	44	500	0.98	0.582	100	0	0	Tautological collapse
	44	1000	1.00	0.002	0	85	0‡	H8=15%
	3-seed mean±std		0.68±0.47		0.238±0.30			500-step
	3-seed mean±std		0.95±0.05		0.16±0.27		0.0±0.0	1000-step
ISR β=0.1								
	42	500	0.81	0.415	16.0	51.9	**17.3**	Mixed H1+H7+H8
	43	500	0.61	0.830	**100**	0	0	H7a=59, H7b=2
	44	500	0.89	0.902	100	0	0	H7 dominant (FN corrected)
	3-seed mean±std		0.77±0.14		0.716±0.26			
ISR β=0.3								
	42	500	0.18	0.143	—	—	—	PR collapse
Random β=0.1								
	42	500	0.79	0.131	0	73.4	22.8	H8=4
	43	500	0.94	0.150	**38.3**	59.6	1.1	H7a=36; H8=1
	44	500	0.89	0.070	0	82.0	6.7	H8=16 (H1∩H8: 6)
	3-seed mean±std		0.87±0.08		0.117±0.042			H7 seed-dep.: 0/38/0%
Random β=0.3								
	42	500	0.48	0.094	8	92	0	H7a=4 (D086); n_{comp} = 9.6
	43	500	0.92	0.180	7.6	89.1	3.3	H7a=7; n_{genuine}=3
	44	500	0.23	0	0	100	0	Pure H1 (n=23)
	3-seed mean±std		0.54±0.35		0.09±0.09			

Table 7: Detailed per-seed results: Prolog (SWI-Prolog 8.4.2). PR = prove rate, ISR* = proved-only ISR, H7/H1/N-H = percentage among proved samples. "—" indicates data not available.

Condition	Seed	PR	ISR*	H7%	H1%	N-H%	Notes
Exec-only							
	42	0.83	0.037	0	0	—	Ground fact enumeration
	43	0.56	0.015	—	—	—	
	44	0.21	0.387	—	—	—	Small-sample (n=21)
	3-seed mean±std		0.53±0.31		0.146±0.21		
Exec+ISR (STA reward)							
	42	0.50	1.000	**100**	0	0	H7 tautological
	43	0.38	0.185	0‡	97.4	0	‡Functional H7[a]
	44	0.97	0.912	0‡	0	0‡	Query-as-fact gaming[b]
	3-seed mean±std		0.62±0.31		0.699±0.45		3/3 Goodhart, 0% non-hacking

[a] Strict H7 classifier reports 0%, but functional analysis shows all 38/38 proved samples are tautological ($n_{\text{necessary}} = 1$). Uppercase pseudo-Prolog syntax bypasses the text-match classifier.

[b] Strict classifier reports Non-hacking = 100%, but 95/97 proved samples have $n_{\text{necessary}} = 1$. The model converts NL to snake_case atoms serving as both fact and query—a third distinct Goodhart pathway.

‡ Strict classifier value is misleading; see footnotes [a]/[b].

Table 8: Per-seed results at β=0.3 (FOL). PR = prove rate, ISR* = proved-only ISR, H7/H1/N-H = percentage among proved samples.

Condition	Seed	PR	ISR*	H7%	H1%	N-H%	Mode
ISR β=0.3							
	42	0.18	0.143	0	77.8	5.6	Reward collapse (n=18); H8=3
	43	0.86	0.295	4.7	68.6	26.7	H1 shortcutting
	44	0.95	0.742	**100**	0	0	Pure H7 meta-Goodhart
	3-seed mean±std		0.66±0.42		0.39±0.31		
Random β=0.3							
	42	0.48	0.094	8	92	0	H7a=4 (D086)
	43	0.92	0.180	7.6	89.1	3.3	H7a=7; n_{genuine}=3
	44	0.23	0	0	100	0	Pure H1 (n=23)
	3-seed mean±std		0.54±0.35		0.09±0.09		